

Name Nitra buseng Setnol^o.

Exercise 2.

SQL Foundations

SQL Aggregate Functions & Operators

Q1 SELECT DISTINCT department
FROM students;

department
IT
HR
Finance.

Q2 SELECT department, AVG(age) AS avg_age
FROM students
GROUP BY department;

Q3

department	avg_age
IT	20.5
HR	22
Finance	23

Q3 SELECT department, COUNT(*) AS student_count
FROM students
GROUP BY department
HAVING COUNT(*) > 1;

department	student_count
IT	2
HR	2

Q4 SELECT * ~~student~~ id, name
FROM Students
WHERE age BETWEEN 21 AND 23;

Student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR.

Q5 SELECT *
FROM students
WHERE department IN ('IT', 'HR')
AND age > 21;

Student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR.

TABLE 02 - COURSES

Q6 SELECT department, SUM(credits) AS total_credits
FROM courses
GROUP BY department
HAVING SUM(credits) > 5;

department	total_credits
IT	11

Q7 SELECT course_id, course_name, department, credits
FROM courses
WHERE NOT (credits = 4);

Course_id	course_name	department	credits
101	SQL Basis	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

Q8 SELECT course_id, course_name, credits
 FROM courses
 ORDER BY credits DESC
 LIMIT 3;

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

Q9 SELECT
 MAX(grade) AS max_grade, MIN(grade) AS
 min_grade, AVG(grade) AS avg_grade
 FROM enrollments;

max_grade	min_grade	avg_grade
90	78	84.6

Q10 SELECT course_id, COUNT(*) AS enrollment
 FROM enrollments
 GROUP BY course_id;

Course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

Q11 SELECT department, SUM(salary) AS total_salary,
 SUM(bonus) AS total_bonus
 FROM salaries
 GROUP BY department;

department	total_salary	total_bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

Q12

```
SELECT department, AVG(salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG(salary) > 55000;
```

department	avg_salary
IT	61000
Finance	70000

Q13 SELECT employee_id, name, salary, bonus,
(salary + bonus) AS total_compensation
FROM salaries
WHERE (salary + bonus) > 60000;

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65000
3	Spice	70000	6000	76000
4	Tyke	62000	5500	67500

Q14 SELECT department, Sum(budget) AS total_budget
AVG(budget) AS avg_budget
FROM projects
GROUP BY department
HAVING AVG(budget) > 70000;

department	total_budget	avg_budget
IT	270000	135000
finance	80000	80000

Q15 Select *

FROM projects

WHERE budget BETWEEN 50000 AND
120000 AND department NOT IN
(Marketing) ;

project_id	Project_name	department	budget
1	AT App	IT	120000
2	payroll system	Finance	80000
5	HR Portal	HR	50000